

# Mansi Maheshwari

+1 (206) 607-7614 | [mmaheshwari@umass.edu](mailto:mmaheshwari@umass.edu) | [LinkedIn](#) | [mansi-maheshwari.com](https://mansi-maheshwari.com)

## EDUCATION

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### Mila – Quebec AI Institute, Polytechnique Montréal

*Ph.D. in Computer Science; Advisor: Sarath Chandar*

Montréal, QC

Sept. 2026 – Present

- Research focus: Continual and Multi-Agent Reinforcement Learning

### University of Massachusetts Amherst

*Master's of Science in Computer Science (GPA 4.0)*

Amherst, MA

Aug. 2024 – May 2026

- Thesis: Continual Reinforcement Learning; Advisor: Bruno Castro da Silva
- Selected as Commencement Speaker for the graduating class [\[video\]](#)

### University of Washington

*Bachelor's of Science in Electrical Engineering*

Seattle, WA

Aug. 2018 – June 2022

- Relevant Coursework: Fundamentals of Optimization and Machine Learning, Signal Processing, Statistical Methods for Science, Embedded Systems

## PUBLICATIONS

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**Maheshwari, M.**, J. C. Raisbeck, and B. Castro da Silva. *AltNet: Addressing the Plasticity–Stability Dilemma in Reinforcement Learning*. Accepted for **oral** presentation at the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS**), 2026. [\[paper\]](#)

**Maheshwari, M.**, B. Castro da Silva, and J. C. Raisbeck. *AltNet: Alternating Network Resets for Plasticity*. In Proceedings of the Conference on Lifelong Learning Agents (**CoLLAs**), 2025. [\[paper\]](#)

## RESEARCH TALKS

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### Invited Talk, Cohere

April 2026

*Presented research on the plasticity-stability dilemma in reinforcement learning [\[video\]](#)*

### Finalist, Three Minute Thesis Competition

March 2025

*Recognized as one of the top 10 students in a university-wide competition, where I presented my research on Continual Reinforcement Learning [\[video\]](#)*

## RESEARCH AND INDUSTRY EXPERIENCE

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### Research Assistant (Continual Reinforcement Learning)

July 2024 – Present

*Autonomous Learning Lab (Advisor: Bruno Castro da Silva), University of Massachusetts*

Amherst, MA

- \* Developed a novel deep reinforcement learning architecture that improves continual learning, enhances sample efficiency, and increases safety in non-stationary environments.
- \* Evaluated agents on robotics control tasks and game environments (DeepMind Control Suite, MuJoCo).
- \* Published as a workshop paper at the Conference on Lifelong Learning Agents (CoLLAs) 2025 and later accepted for **oral** at the International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2026.

### Artificial Intelligence Research and Development Intern

May 2025 – Aug 2025

*Perception Team, CNH Industrial*

Scottsdale, AZ

- \* Led the design and development of an efficient, scalable vision architecture unifying object detection and segmentation through a **transformer** based YOLO multihead model for autonomous vehicles.
- \* Reduced parameters by ~43%, cut inference cost by ~30%, and outperformed baselines by ~5-7%.
- \* Developed and documented a multi-task architecture that would allow new vision tasks to be added with minimal compute overhead, establishing a scalable foundation for future vision capabilities.

## Software Engineer

July 2022 – July 2024

*Nordstrom*

*Seattle, WA*

- \* Optimized workflow by automating multiple engineering tasks (Java) in distributed systems.
- \* Achieved 80% test coverage for large-scale data integrity through JUnit Integration Tests for multiple projects.
- \* Led end-to-end development (requirements gathering, design discussions, code reviews, testing, and deployment) of a feature to stop awarding points for alcohol purchase.

## PROJECTS

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### Human Following Robot, Autonomous Cinematography

Feb 2025 – May 2025

*Robotics Course*

- \* Designed and implemented a mobile robot to autonomously follow and film a moving subject.
- \* Integrating perception (YOLOv7 for human detection and tracking), path planning (DWA-based trajectory generation), and real-time control (PID-based actuation)

### Multi-Modal Conversational Recommender System (LLMs)

Aug 2024 – Dec 2024

*Neural Networks Course*

- \* Designed a Transformer-based sequential movie recommendations on MovieLens 1M with causal masking. Combines text (natural-language queries and movie titles via a pre-trained **LLM**-based sentence encoder) and sequential/collaborative signals (user ID and watch history).
- \* Evaluated against Markov chain, SVD collaborative filtering, and popularity baselines using NDCG@K, Precision@K, and Hit Rate@K; the Transformer outperformed all baselines.

## TEACHING EXPERIENCE

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### Instructor, Fundamentals of Artificial Intelligence

May 2025 – July 2025, Feb 2026 – May 2026

*University of Washington*

*Remote*

- \* Developed the curriculum and designed accessible, visually engaging slide decks and coding exercises.
- \* Instructed a 10-day course, consisting of 3-hour lectures, introducing high school students to core AI concepts including machine learning, deep learning, computer vision, large language models, and ethical AI.
- \* Guided students in completing final projects synthesizing learned skills.
- \* Fostered an inclusive and interactive classroom environment through live polls, quizzes, reflection activities, and curated videos showcasing real-world AI applications to enhance engagement and conceptual understanding.

### Subject Matter Expert, AI Curriculum (High School)

Sept 2025 – Present

*iCEV Multimedia*

*Remote*

- \* Designed curriculum and reviewed lesson plans to ensure accurate, scaffolded content for an introductory AI textbook for high school students.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++, Java, JavaScript, R, MATLAB

**DL Frameworks:** PyTorch, TensorFlow, JAX

**Systems:** CUDA, TensorRT, ROS, Linux

**Domains:** Deep/Reinforcement Learning, LLMs, Transformers, Multi-Task Learning, Representation Learning

## ACADEMIC SERVICE

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**Reviewer,** Reinforcement Learning Conference (RLC)

2026